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Cognitive Function Assessment in Mandibular Prognathism: A Pilot Study Investigating Relationship Between Chewing-Induced Brain Blood Flow and Eye-Tracking Performance

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Article

Keywords:

Posted Date: November 5th, 2025

DOI: https://doi.org/10.21203/rs.3.rs-7794817/v1

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Additional Declarations: No competing interests reported.

Version of Record: A version of this preprint was published at Scientific Reports on January 16th, 2026. See the published version at https://doi.org/10.1038/s41598-026-35964-x.

Abstract

Despite established links between masticatory dysfunction and cognitive impairment, cognitive function in dentofacial deformity patients remains unexplored. This study represents the first comprehensive cognitive assessment in mandibular prognathism (MP) patients.

Brain blood flow (BBF) during chewing was measured using functional near-infrared spectroscopy in MP patients versus normal occlusion controls (NORM, n = 17). Cognitive function using eye-tracking technology (Mirudake) in MP patients (n = 44) was compared to healthy controls (HC, n = 59), with assessing six cognitive domains.

MP patients demonstrated significantly reduced chewing-induced BBF in bilateral inferior frontal gyrus compared to NORM. Global cognitive performance showed no significant difference between MP and HC groups (p = 0.48). However, positive correlations existed between BBF and cognitive performance domains including global performance and memory. ROC analysis using pooled bilateral BBF data (n = 88) revealed modest diagnostic potential for cognitive assessment (AUC = 0.657, 95% CI: 0.431–0.865). The optimal threshold yielded 62.5% sensitivity, 73.8% specificity, and 72.7% overall accuracy for detecting cognitive impairment.

This first systematic cognitive evaluation of MP patients revealed no significant global cognitive impairment despite confirmed reductions in masticatory-induced brain activation. The observed BBF-cognition correlations suggest BBF could serve as a novel biomarker for cognitive function assessment, warranting further investigation through longitudinal studies examining orthognathic surgery's potential cognitive effects.

Introduction

Dentofacial deformity, characterized by significant skeletal discrepancies in the maxillofacial region, affects approximately 1–5% of the general population, with prevalence varying considerably across different ethnic groups and diagnostic criteria [1]. These skeletal abnormalities, including mandibular prognathism, retrognathism, and facial asymmetry, result not only in aesthetic concerns but also in substantial functional impairments of the stomatognathic system. Multiple studies have demonstrated that patients with dentofacial deformities exhibit significantly reduced masticatory performance compared to individuals with normal occlusion [2, 3]. The masticatory efficiency in these patients has been reported to be reduced by 30–50%, with difficulties in food comminution and bolus formation [4, 5].

The bidirectional relationship between masticatory function and cognitive performance is supported by converging neuroimaging, epidemiological, and interventional evidence, establishing mastication as a critical modulator of brain health [6, 7]. Epidemiological studies demonstrate that tooth loss is associated with increased cognitive impairment risk in older adults, independent of socioeconomic and health factors [8]. Similarly, prospective cohort study of Japanese older adults revealed that self-reported poor dental health status was associated with increased risk of dementia onset over a 4-year follow-up period [9]. Chewing activates brain regions including sensorimotor cortex, prefrontal cortex, hippocampus, and cerebellum, with fMRI showing 15–25% increased activity [10, 11]. Moreover, positron emission tomography studies demonstrate that chewing increases glucose metabolism in brain regions crucial for executive function and attention [12]. We previously reported that jaw deformity such as mandibular prognathism (MP) [13] and

skeletal open bite [14] significantly attenuates chewing-induced brain blood flow (BBF) at an area reported to play an important role in cognitive function [15, 16], suggesting possible retardation by malocclusion in cognitive function. Studies reveal 1.23–1.87 times higher dementia risk with poor masticatory function after adjusting for confounders [17, 18]. Masticatory training improved working memory and processing speed in older adults with mild cognitive impairment [19]. Each functional tooth pair reduces dementia risk by ~ 3.1%, indicating a quantifiable oral-cognitive relationship [20]. These findings collectively position masticatory function not merely as a correlation but as a potentially modifiable determinant of cognitive health.

Despite established links between masticatory dysfunction and cognitive impairment, there remains a knowledge gap regarding cognitive function in dentofacial deformity patients, who have severe chronic masticatory impairment from developmental rather than age-related origins [21, 22]. Systematic literature search reveals that while numerous studies examine functional and psychosocial impacts of dentofacial deformities, investigations of cognitive function in this population are absent, representing a significant oversight given these patients experience lifelong masticatory efficiency reductions [23]. This research gap is concerning as dentofacial deformity patients differ fundamentally from elderly populations: dysfunction occurs during critical neurodevelopmental periods, is biomechanical rather than purely dental, and affects younger individuals whose cognitive reserve and neuroplasticity may respond differently to chronic masticatory impairment.

Eye-tracking technology has emerged as a validated and sensitive biomarker for cognitive function assessment [24], with recent systematic reviews confirming its superiority over traditional screening tools [25, 26]. Meta-analysis by Opwonya et al. of 35 studies showed saccadic eye movement parameters distinguished cognitive impairment with pooled sensitivity of 0.84 and specificity of 0.81, outperforming conventional cognitive screening instruments [27]. Semmelmann and Weigelt demonstrated that webcam-based eye-tracking achieves high correlation with lab equipment and sufficient resolution for detecting millisecond attention changes [28]. Validation studies show low-cost eye-tracking systems match research equipment performance, with test-retest reliability > 0.85 for cognitive assessments [29]. Kaufmann et al. found eye-tracking detected cognitive deficits with 89% accuracy (vs 67% for traditional tests), reducing assessment time by 40% and eliminating ceiling effects [30]. This approach could reveal cognitive implications of masticatory dysfunction in dentofacial deformity patients, uncovering unrecognized disease burden to inform treatment strategies.

Given this knowledge gap and validated eye-tracking methodology, this study aims to evaluate cognitive function in dentofacial deformity patients using eye-tracking-based cognitive assessments compared to healthy controls. Specifically, we aim to quantify differences in multiple cognitive domains, thereby establishing the first comprehensive cognitive profile of the patient with malocclusion, specifically, MP. This investigation will not only fill a significant gap in our understanding of the systemic impacts of dentofacial deformities but also provide crucial evidence for optimizing treatment timing and developing targeted cognitive rehabilitation strategies for this patient population, potentially establishing masticatory function restoration as a modifiable factor for cognitive health maintenance across the lifespan.

Materials and methods

Ethical Approval

The present investigation received ethical clearance from the Institutional Review Board of Tsurumi University School of Dental Medicine under approval numbers 1316 and 121005. The study protocol adhered strictly to the ethical principles outlined in the Declaration of Helsinki regarding human research participation. Prior to enrollment, comprehensive written informed consent was secured from all study participants following detailed explanation of the research objectives, procedures, and potential risks. The study design and reporting were conducted in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement guidelines to ensure methodological rigor and transparency in human observational research.

Sample Size Calculation

Sample size calculation was conducted prior to data collection using G*Power 3.1.9.4 software (Universität Kiel, Kiel, Germany). The analysis was based on an anticipated effect size of 0.8, with alpha set at 0.05 and statistical power at 0.8. This yielded a minimum required sample size of 54 participants for sum of two groups, which matched our final sample size of (61 for BBF comparison, 103 for cognitive function comparison) participants.

Subjects

The experimental group consisted with the patients with MP (MP group, n = 44), comprising 21 men and 23 women (age: 25.0 ± 10.4). The MP group comprised patients who presented to our hospital for initial consultation regarding surgical orthodontic treatment. Participants were required to meet the following inclusion criteria:

-Absence of congenital abnormalities

-Complete permanent dentition excluding third molars

-Presence of anterior crossbite

-Skeletal and dental midline deviation ≤1.0 mm from the facial midline

-Absence of functional symptoms, including temporomandibular joint disorders

-No previous history of orthodontic treatment

-Angle Class III molar relationship with mandibular first molar positioned anteriorly relative to the maxillary first molar

-Skeletal Class III intermaxillary relationship as determined by ANB angle

Exclusion criteria included systemic diseases or concurrent medication use that could potentially influence cerebrovascular regulation or neuromuscular function.

All participants completed a comprehensive health questionnaire and underwent a detailed clinical interview to verify adherence to the specified inclusion and exclusion criteria prior to enrollment.

Control group for BBF comparison

A control group (NORM group) comprising 17 participants (4 males, 13 females; mean age: 19.4 ± 0.6 years) was established based on the following inclusion criteria:

-Complete dentition excluding third molars (no missing teeth other than wisdom teeth)

-Normal overjet and overbite measurements (2-4 mm for each parameter, indicating appropriate occlusion in the anterior region)

-Skeletal and dental midline deviation ≤1.0 mm from the facial midline (absence of significant lateral jaw deviation)

-Absence of functional disorders, including temporomandibular joint dysfunction

-No prior history of orthodontic intervention

-Angle Class I molar relationship (indicating favorable anteroposterior relationship between maxillary and mandibular first molars)

-Normal intermaxillary relationships, as evidenced by cephalometric measurements (ANB angle: 3.1 ± 2.5°; mandibular plane to Frankfort horizontal plane angle: 26.6 ± 6.7°)

Control group for cognitive function comparison

Data from 59 healthy participants (29 males, 30 females, age: 44.8 ± 3.0) were provided by Ai-BrainScience Inc. as healthy control (HC) group data (unpublished data). These data were collected under the approval of the Ethics Committee of Yoga Allergy Clinic (approval number: AI11001XX6).

The following inclusion criteria were applied for control group selection:

-Age 40 years or older and younger than 50 years at the time of consent acquisition

-No history or comorbidity of dementia, active neurological diseases, or psychiatric disorders at the time of consent acquisition

Participants meeting any of the following criteria were excluded from the study:

-Individuals with ophthalmological conditions that significantly impair eye-tracking detection (e.g., strabismus, cataracts resulting in insufficient visual acuity, or visual field defects due to glaucoma or diabetes), rendering eye-tracking detection impossible. Participants using the following vision correction devices were required to perform testing without these devices: 1) Sunglasses with insufficient brightness or difficulty in color recognition. 2) Eyeglass frames that are extremely small or excessively thick. 3) High myopia (-6.25 diopters or greater) with significant refractive correction lenses that impede eye-tracking. 4) Post-cataract surgery correction lenses (+13 diopters approximately) with significant refractive correction that impedes eye-tracking. 5) Bifocal or multifocal lenses with convex lenses for near vision in the lower portion.

-Neurological disorders with motor impairments that clearly compromise the ability to maintain posture during measurement

-Individuals who had undergone the measurement of cognitive function by Mirudake within 3 months prior to this study

-Other conditions including nystagmus or trauma that prevent appropriate measurement of gaze data, or individuals unable to maintain posture during testing

BBF Measurement

BBF was assessed using functional near-infrared spectroscopy (fNIRS; OEG-16 device; Spectratech, Tokyo, Japan), which quantifies oxygenated hemoglobin (oxy-Hb) concentrations in the prefrontal cortex through 16 channels. Detailed methodological procedures for BBF measurements have been previously documented [13,14]. Of the 16 available channels, channels 1-4 and 13-16, which encompass the inferior frontal gyrus, were selected for analysis based on established evidence demonstrating the critical role of this brain region in cognitive processing [15,16].

Prior to experimental procedures, measurement calibration was performed and demonstrated high reproducibility of BBF responses during chewing tasks, with intraclass correlation coefficients (ICC) exceeding 0.7 after a one-week interval [13]. To ensure consistent fNIRS data acquisition across all experimental trials, standardized probe placement protocols were implemented using conventional headgear systems. Participants received explicit instructions to maintain stable head positioning throughout the measurement period to minimize motion-related artifacts. All measurements were conducted in a controlled, quiet environment with reduced external stimuli to eliminate potential confounding variables.

BBF were measured in NORM and MP groups.

Task Setting for fNIRS

A simple block design was employed, consisting of baseline rest periods and experimental conditions. The positive control condition comprised a cognitive arithmetic task, wherein participants performed verbal serial subtraction of 7 from 100. The experimental condition involved masticating soft CRT paraffin (Ivoclar Vivadent, Tokyo, Japan) using their preferred masticatory side. Each condition was executed for approximately 60 seconds, with participants maintaining their natural masticatory rhythm. To ensure adequate consistency in chewing patterns, all sessions were monitored by the investigator, and participants received instructions to utilize their preferred masticatory side consistently across all trials. Although individual masticatory frequencies were not rigidly standardized, participants were instructed to sustain a comfortable and consistent rhythm throughout each condition to reduce intra-individual measurement variability. Following each experimental condition, a minimum 30-second rest period was implemented to permit oxy-Hb concentrations to return to baseline levels. Baseline recovery was verified through continuous oxy-Hb signal monitoring prior to initiating subsequent conditions.

fNIRS Data Analyses

Comprehensive methodological details for fNIRS data analysis have been previously documented [13,14]. In summary, peak oxy-Hb values within each channel during respective experimental conditions constituted the primary analytical focus of this investigation. Peak oxy-Hb values were computed for each condition and subsequently subjected to inter-group comparative analysis and correlation analysis against cognitive function

domains. For quantification of peak oxy-Hb concentrations, the median value derived from the maximum concentrations recorded across channels 1-4 and channels 13-16 was designated as the representative metric for each participant.

Measurement of cognitive function (Mirudake measurement).

We measured the cognitive function of the subjects with an eye-tracking application, Mirudake (Ai-BrainScience Inc., Osaka, Japan). Briefly, Mirudake is an eye-tracking application designed for iPadOS 12.1 or later that utilizes ARKit-based technology for gaze measurement. We used an iPad Pro (12.9-inch, 264 ppi) mounted inverted on a stand with the front camera capturing participants' eye movements. The device was positioned 51 cm from participants at a 90-degree angle to the table.

The measurement protocol begins with face position adjustment, where participants align themselves with an on-screen guide before raw gaze data collection commences. To ensure accurate mapping of gaze data to screen coordinates, Mirudake employs a five-point calibration system followed by an accuracy verification phase. To maintain sufficient precision for cognitive assessment, the system requires that the Euclidean distance between on-screen targets and estimated gaze positions falls within predetermined thresholds during accuracy verification. This study implemented thresholds of either 200 points (38 mm) or 300 points (57 mm), with prior validation confirming no significant differences in cognitive function scores between these settings. When participants fail the accuracy check, the system automatically reinitiated the five-point calibration sequence. This calibration-verification cycle continues until either successful completion or three consecutive failures, at which point the application displays a warning and returns to the initial face position adjustment screen. Participants who failed to achieve acceptable accuracy after four complete cycles (comprising up to 12 calibration and verification attempts) were excluded from the study.

The assessment evaluated six primary cognitive domains: memory, executive function, attention, visuospatial cognition, language, and orientation. Global cognitive performance score was derived by averaging performance scores from the six assessed cognitive domains.

Cognitive functions were measured in MP group, and the data were compared to HC data provided by Ai-BrainScience Inc.

Statistical Analyses

Statistical computations described below were executed using SPSS Statistics software (version 27.0, IBM, Tokyo, Japan).

Comparison between groups

For between-group comparisons, we utilized a dual analytical approach combining traditional non-parametric methods (Mann–Whitney U test) with bootstrap resampling techniques to ensure comprehensive and reliable interpretation of results. Given the non-normal distribution characteristics of our dataset and the need for outlier-resistant analysis, the Mann–Whitney U test served as our primary comparative method. To enhance the robustness of our statistical inferences, particularly considering our moderate sample size, we supplemented this approach with bootstrap analysis involving 1000 resampling iterations. This dual methodology provided both conventional hypothesis testing (with statistical significance set at $p < 0.05$) and detailed assessment of

effect reliability through bootstrap-derived confidence intervals (CI). We considered between-group differences statistically meaningful when the 95% CI for the median difference excluded zero, demonstrating consistent directional effects.

Equivalence test

Equivalence testing was performed with the setting of equivalence margins using distribution-based methodology [31]. Two margins were established: 1. conservative margin: ±0.2 ´ pooled standard deviation (SD), 2. standard margin: ±0.5 ´ pooled SD. Equivalence was assessed using the Two One-Sided Tests (TOST) procedure, which tests whether the group difference falls within the predetermined equivalence margins. The TOST procedure consists of:

1. Lower bound test: Tests whether the HC group mean is not substantially lower than the MP group mean
2. Upper bound test: Tests whether the HC group mean is not substantially higher than the MP group mean

Statistical equivalence was declared if both one-sided tests yielded p-values < 0.05. Additionally, the 90% CI for the group difference was examined. Equivalence was supported if the entire interval fell within the equivalence margins.

Correlation analysis

For examining relationships between variables, we conducted correlation analyses using both Spearman's rank correlation and Pearson correlation methods, each enhanced with bootstrap resampling (1000 iterations) to improve estimate stability.

Evaluation of the diagnostic performance of BBF measurements in cognitive assessment

Evaluation of the diagnostic performance of BBF measurements in cognitive assessment were performed as follows.

1. To maximize statistical power and account for potential bilateral variations, data from both left and right sides were pooled, resulting in a total sample size of 88 measurements (n = 88) for the subsequent analyses.
2. The threshold value for classification was established using cognitive function measurements obtained through the Mirudake assessment tool. The Mirudake comprehensive scores were converted to Mini-Mental State Examination (MMSE) equivalent values using established conversion tables [32]. The threshold was set as the global cognitive performance score 50, which corresponding to the lower limit of the normal range in MMSE (i.e., MMSE ≥ 26), representing the boundary between normal cognitive function and cognitive impairment.
3. Receiver operating characteristic (ROC) curve analysis was performed to evaluate the diagnostic performance of BBF measurements in discriminating between cases below and above the predetermined threshold. The area under the ROC curve (AUC) was calculated as the primary measure of discriminative ability.
4. CIs for the AUC were calculated using the bootstrap method with 1000 resampling iterations. This non-parametric approach was selected due to the relatively small number of positive cases (n = 4), as bootstrap

methods provide more robust CI estimates compared to normal approximation methods in small sample studies. Bootstrap resampling was performed with replacement from the original dataset, and the 95% CI was determined using the 2.5th and 97.5th percentiles of the bootstrap distribution.

5. The optimal threshold for diagnostic classification was determined using Youden's index (sensitivity + specificity - 1), which maximizes the sum of sensitivity and specificity. Additional performance metrics, including sensitivity, specificity, positive predictive value, negative predictive value, and overall accuracy, were calculated at the optimal threshold.

Results

Baseline characteristics.

Regarding the sex ratio, no statistically significant differences were observed between groups. However, significant differences in mean age were found among the three groups (NORM, MP, and HC; $p < 0.05$). Therefore, prior to the following analysis, we examined whether age influences BBF and cognitive function within each group. BBF exhibited no correlation with age in either group. Cognitive function also showed no correlation with age in both the HC and MP group. These findings indicating no correlation between age and either BBF or cognitive function suggest that despite the observed age differences between groups, the influence of age is limited. Consequently, the subsequent between-group comparisons can be considered to reflect differences arising from the distinct characteristics of each group rather than age-related confounding effects.

Mandibular prognathism attenuates chewing-induced BBF.

To examine the effects of mandibular prognathism on chewing-induced BBF, we measured regional BBF in the bilateral inferior frontal gyrus during paraffin chewing tasks. Statistical analysis revealed significant differences in chewing-induced BBF between NORM and MP in both hemispheres. In the right inferior frontal gyrus, NORM demonstrated significantly higher percent induction of BBF compared to MP (median [IQR]: NORM 57% [20-80%] vs. MP 23% [10-45%]; $p < 0.05$). Similarly, in the left inferior frontal gyrus, NORM showed significantly greater activation than MP (median [IQR]: NORM 50% [15-75%] vs. MP 33% [5-60%]; $p < 0.05$).

Both Mann-Whitney U test and bootstrap analysis confirmed the statistical significance of these bilateral differences, demonstrating robust evidence for attenuated chewing-induced BBF in patients with MP. This observed pattern was consistent across both hemispheres, suggesting a generalized reduction in cortical activation during mastication in MP.

These findings replicate our previous report [13] and provide further evidence that MP significantly impacts the neural mechanisms underlying masticatory function, as reflected by reduced BBF responses in the inferior frontal gyrus during chewing activities.

Comparison of cognitive function scores between HC and MP.

To assess whether MP gives any influence on cognitive function, we compared Mirudake scores in both HC and MP. Statistical analysis revealed no significant difference in global cognitive performance score ($p = 0.4776$, 95% CI: [-2.8, 5.9], Figure 2A). The median of global cognitive performance scores was comparable between

groups, with HC showing a median score of approximately 73 and MP showing a median score of approximately 70. Despite the lack of statistical significance, descriptive analysis revealed a numerical tendency for lower global cognitive performance score in MP compared to HC.

To further clarify whether there is any difference between HC and MP in global cognitive performance score, equivalence analysis was performed (Table 1). Conservative margin did not demonstrate statistical equivalence (lower bound p = 0.032, upper bound p = 0.377) in the TOST procedure. The 90% CI exceeded the conservative equivalence margin (±2.2 points), as the upper bound (5.15) was greater than +2.2. On the other hand, standard margin (±5.5 points) confirmed statistical equivalence (lower bound p = 0.001, upper bound p = 0.035). The entire 90% CI fell within the standard equivalence margins. This result suggests that while MP may exhibit a subtle tendency toward reduced cognitive function as measured by Mirudake scores, this difference does not reach statistical significance and may not represent a clinically meaningful impairment in cognitive function assessment.

We further compared each of the six primary cognitive domains (Figures 2B to 2G). Almost all domains exhibited no statistical difference between the groups. However, orientation score shows higher value in MP as compared to that in HC, with statistical significance (Figure 2G).

Correlation analysis between BBF and cognitive functions.

We then performed correlation analysis between BBF and cognitive functions (Table 2). BBF at both the right and the left inferior frontal gyrus exhibited some positive correlations to cognitive function scores, such as global cognitive performance and memory. Furthermore, visuospatial cognition score indicates positive correlation to BBF at the right side, though executive function score indicates positive correlation to BBF at the left side. Higher BBF values were associated with higher cognitive performance scores, while lower BBF values were associated with lower cognitive performance scores.

Correlation analysis among cognitive function parameters within MP.

Correlation analyses between cognitive function parameters were conducted using both Spearman and Pearson with bootstrap resampling methods. Both analytical approaches demonstrated similar overall patterns of associations among the cognitive domains (Figure 3). The strongest correlation was observed between the global cognitive performance score and memory function (Spearman r = 0.74, Pearson r = 0.75). The global cognitive performance score also showed moderate correlations with executive function (Spearman r = 0.62, Pearson r = 0.66) and language (Spearman r = 0.49, Pearson r = 0.58). Among the individual cognitive domains, memory demonstrated a moderate correlation with language (Spearman r = 0.55, Pearson r = 0.46). Executive function showed weak to moderate correlations with orientation in both analyses (Spearman r = 0.39, Pearson r = 0.36).

Notably, correlations involving visuospatial cognition showed discrepancies between the two analytical methods, with several correlations reaching significance in the Pearson analysis (e.g., visuospatial cognition-global cognitive performance score: r = 0.61) but not in the Spearman analysis (r = 0.40). Many inter-domain correlations, particularly those involving attention, visuospatial cognition, and orientation, did not reach statistical significance in either analysis.

Evaluation of the diagnostic performance of BBF measurements in cognitive assessment.

ROC curve analysis was performed to evaluate the discriminative ability of BBF measurements for cognitive status classification (Figure 4). The area under the ROC curve (AUC) was 0.657 (95% CI: 0.431-0.865, bootstrap method with 1000 resampling iterations). This AUC value indicates discriminative ability above chance level (AUC = 0.5), suggesting that BBF measurements have potential utility as a diagnostic biomarker for cognitive assessment. Despite the lack of statistical significance, the effect size analysis revealed a moderate effect (Cohen's d = 0.510, 95% CI: -0.221 to 1.240), suggesting clinically meaningful differences between groups. The wide CI reflects the uncertainty inherent in small sample analyses, with the interval encompassing both null and substantial effects. Power analysis indicated that approximately 61 subjects per group would be required to detect the observed effect size with 80% statistical power at α = 0.05, compared to the current 8 subjects in the below-threshold group. The estimated statistical power of the current study was approximately 82%, suggesting adequate power despite the non-significant result.

Threshold performance analysis revealed multiple viable cutoff values (Table 3), with Youden's index ranging from 0.109 to 0.363, indicating varying trade-offs between sensitivity and specificity across different threshold selections.

The optimal diagnostic threshold was determined using Youden's index, which maximizes the sum of sensitivity and specificity (Table 4). The optimal BBF threshold was identified as ≤0.068174, yielding the following performance characteristics. Sensitivity: 62.5% (95% CI: 29.0-96.0%), Specificity: 73.8% (95% CI: 64.1-83.4%), Positive predictive value: 19.2% (95% CI: 4.1-34.4%), Negative predictive value: 95.2% (95% CI: 89.8-100.0%), Overall accuracy: 72.7% (95% CI: 63.4-82.0%), and Area Under Curve: 0.657 (95% CI: 0.431-0.865).

Discussion

This study represents the first comprehensive investigation of cognitive function in dentofacial deformity patients, addressing a critical knowledge gap regarding systemic impacts of severe masticatory dysfunction. Primary findings demonstrate that MP patients did not exhibit statistically significant cognitive impairment compared to HC. Equivalence analysis revealed that groups existed at the threshold between equivalence and non-equivalence, with conclusions dependent on margin stringency. Using conservative criteria (±0.2 SD), equivalence was not demonstrated, whereas standard criteria (±0.5 SD) supported equivalence. The observed mean difference of 1.56 points, while statistically non-significant (p = 0.555), represents 71% of the conservative margin and 28% of the standard margin, placing groups in an intermediate zone. The negligible effect size (Cohen's d = 0.142) supports that any between-group difference lacks practical significance. These borderline results emphasize the critical importance of a priori equivalence margin specification and suggest that larger sample sizes may provide more definitive evidence.

Significantly, chewing-induced BBF was reduced in MP patients' bilateral inferior frontal gyrus regions. Correlation analyses revealed positive associations between BBF and multiple cognitive domains, including global performance and memory, establishing a neurobiological link between masticatory-induced cerebral activation and cognitive function. These findings provide first empirical evidence for altered neurocognitive processing patterns in dentofacial deformity patients, despite absence of clinically detectable cognitive impairment at the group level.

This study demonstrates that BBF measurements exhibit modest diagnostic performance for cognitive assessment, with an AUC of 0.657 (95% CI: 0.431-0.865). According to established criteria, AUC values are interpreted as: 0.90-1.0 (excellent), 0.80-0.89 (good), 0.70-0.79 (fair), 0.60-0.69 (poor), and 0.50-0.59 (fail) [33]. Our AUC of 0.657 places BBF in the "poor" category, yet above the threshold for potential clinical utility. Many clinically useful biomarkers demonstrate AUC values in the 0.6-0.8 range, particularly in complex conditions where multiple factors contribute to outcomes [34]. The primary constraint is the small number of below-threshold cases (n=8), resulting in wide CIs. Power analysis indicates approximately 60 subjects per group would be required for adequate statistical power. Despite these limitations, the moderate effect size (Cohen's d = 0.510) suggests clinically meaningful differences between groups, supporting BBF's biological plausibility as a cognitive biomarker. While BBF alone may not serve as a standalone diagnostic test, it shows potential value as a screening tool or component of multimodal approaches. Threshold analysis revealed multiple viable cutoff options (Youden's index: 0.109-0.363), providing flexibility for clinical implementation. The MP subjects were relatively young without many pathological subjects with cognitive decline. Therefore, we investigated detection of subjects with values below Mirudake conversion values, corresponding to MMSE lower normal limits. Future studies require large-scale populations including participants with cognitive impairments.

The observed reduction in chewing-induced brain blood flow in mandibular prognathism patients results from impaired stomatognathic function and reduced masticatory sensory input. Our investigations demonstrated that occlusal contact area and jaw movement parameters significantly correlate with masticatory-induced BBF responses [13,14]. These findings suggest that the compromised occlusal function characteristic of dentofacial deformities directly impacts neural activation patterns during mastication, as evidenced by reduced periodontal mechanoreceptor stimulation and diminished masticatory muscle activity [13,14]. As mastication induces trigeminal sensory input [35], the mechanistic pathway likely involves malocclusion-induced reduction in effective masticatory stimulation leading to decreased trigeminal sensory input, subsequently attenuating ascending arousal signals that normally enhance prefrontal cortical blood flow during chewing. While our findings demonstrate significant correlations between BBF measures and cognitive performance domains, suggesting a neurobiological link between masticatory dysfunction and cognitive processing, the cross-sectional design precludes definitive causal inferences. The hypothetical mechanism whereby reduced masticatory stimulation leads to decreased BBF, potentially impacting cognitive function, represents a plausible neurobiological pathway warranting investigation through longitudinal and interventional studies to establish temporal relationships and causal directionality.

Present findings should be interpreted within established research linking masticatory dysfunction to cognitive impairment. Dintica et al.'s 22-year study of 544 adults showed poor masticatory ability accelerated spatial and fluid cognitive decline after age 65 [36]. Similarly, Sta Maria et al.'s systematic review and meta-analysis of studies from 2000-2022 confirmed that older adults with masticatory dysfunction exhibited increased risk of cognitive impairment, with consistent evidence across multiple cognitive domains [37]. However, research investigating mastication-cognition relationships in younger adult populations, especially patient with malocclusion, remains notably limited, with most studies focusing on acute effects of gum chewing [38] rather than chronic masticatory dysfunction. Few young adult studies, like Kim et al.'s investigation of chewing effects on brain antioxidants and cognition, examined temporary enhancement from masticatory stimulation rather than chronic impairment consequences [39]. Importantly, Sta Maria et al. found gum chewing reduced cognitive processing speed in young adults, suggesting age-dependent mastication-cognition relationships [37]. This is

the first systematic cognitive evaluation of dentofacial deformity patients, who have developmental rather than age-related masticatory dysfunction. This distinction is crucial, as dentofacial deformity patients experience biomechanical masticatory impairment [40], potentially producing different cognitive impacts compared to elderly populations with acquired masticatory dysfunction. The absence of significant cognitive differences in our young adult MP population, despite demonstrable BBF reductions, may reflect age-related differences in neuroplasticity and cognitive reserve that warrant further investigation

MP patients showed significantly higher orientation scores than controls, contrasting with reduced performance trends in other cognitive domains. This counterintuitive result may reflect compensatory enhancement of spatial-temporal awareness in response to altered oral sensory feedback. First, since trigeminal sensory pathways connect to parietal cortex, which has reciprocal hippocampal connections for spatial processing [41,42] , chronic trigeminal alterations might indirectly influence spatial cognitive abilities. Second, MP's developmental nature during critical brain maturation [43,44] may facilitate compensatory reorganization that enhances orientation functions while compromising others. Third, eye-tracking orientation assessment may capture different processes than traditional tasks[45], potentially reflecting enhanced visual-spatial strategies compensating for altered orofacial proprioception. MP patients' experience with facial asymmetry may enhance spatial orientation performance. However, this requires cautious interpretation as it may reflect artifacts or sample characteristics. Further investigation with diverse paradigms is needed, given absent similar findings in previous research.

Several limitations must be acknowledged. First, the cross-sectional design precludes causal relationship determination, as temporal precedence cannot be established. Whether BBF reductions preceded, followed, or developed with cognitive changes remains unclear, limiting causative mechanism inference and preventing capture of developmental trajectories or adaptive changes in MP patients. Second, modest sample sizes (MP: n=44; NORM: n=17; HC: n=59) may have limited statistical power to detect subtle cognitive differences. Post-hoc power analysis revealed insufficient power (9.1%) to detect the observed effect size, increasing Type II error risk. While no significant difference was found between HC and MP groups, this null finding cannot be interpreted as equivalent given low statistical power. Future studies should employ larger samples to adequately test group differences or equivalence. To achieve 80% power for detecting small-to-medium effects (Cohen's d = 0.3), approximately 176 participants per group would be required. Alternatively, if the observed negligible effect represents true population difference, results suggest groups may be practically equivalent, warranting formal equivalence testing with appropriate sample sizes. Third, potential confounders include significant age differences between groups, selection bias from clinical MP patient recruitment representing more severe cases, and limited validation of eye-tracking cognitive assessment [45] against neuropsychological batteries in dentofacial deformity populations. Finally, unmeasured variables including socioeconomic status, education, appearance-related anxiety, or sleep quality may have influenced cognitive performance without systematic control.

While no significant cognitive impairments were found in MP patients, correlations between BBF and cognitive performance suggest future research directions. The potential relationship between orthognathic surgery and cognitive enhancement warrants investigation, as benefits may extend beyond aesthetic/functional outcomes. However, rigorous validation through intervention studies is required. Several critical research directions emerge. Longitudinal studies comparing cognitive function before/after orthognathic surgery; large-scale cross-sectional investigations with enhanced power; mechanistic neuroimaging studies elucidating pathways

between masticatory dysfunction and cognition; randomized controlled trials of surgery's cognitive effects; and intervention studies exploring non-surgical masticatory rehabilitation. These findings suggest investigating orthognathic surgery's potential cognitive effects, though validation requires prospective intervention research with appropriate controls.

In conclusion, this study provides the first assessment of cognitive function in MP patients, revealing no significant global cognitive impairment despite confirmed reductions in masticatory-induced BBF. The observed correlations between BBF and cognitive performance suggest potential neurobiological relationships that warrant further investigation. Also, this pilot study establishes preliminary evidence for BBF's diagnostic potential in cognitive assessment. While the modest AUC value and sample size limitations prevent definitive clinical recommendations, the above-chance discriminative ability, moderate effect size, and methodological rigor justify continued investigation. These findings support the feasibility of larger validation studies that could establish BBF's role in comprehensive cognitive healthcare approaches. While these preliminary findings do not establish causality, they provide empirical foundation for future longitudinal studies examining whether orthognathic surgery influences cognitive outcomes. Understanding the broader systemic implications of dentofacial deformities remains an important area for continued research in oral and maxillofacial medicine.

Declarations

Funding:

This work was supported in part by Grants-in-Aid for Scientific Research from the Japan Society for the Promotion of Science (23K09449 and 24K13186).

Author Contribution

Y.I.: Acquisition and analysis of data. Revised the manuscript.H.K.: Conception and design of this work. Interpretation of data. Have drafted the manuscript.C.K.: Design of this work. Acquisition and analysis of data. Revised the manuscript.S.T.: Acquisition of data. Revised the manuscript.M.K.: Conception and design of this work. Revised the manuscript.H.T.: Revised the manuscript.

Acknowledgement

The authors thank Ai-BrainScience Inc. (Osaka, Japan) for generously providing cognitive function data from healthy control participants. This work was supported in part by Grants-in-Aid for Scientific Research from the Japan Society for the Promotion of Science (23K09449 and 24K13186).

Data Availability

The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request.

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Tables

Tables 1 to 4 are available in the Supplementary Files section.

Figures

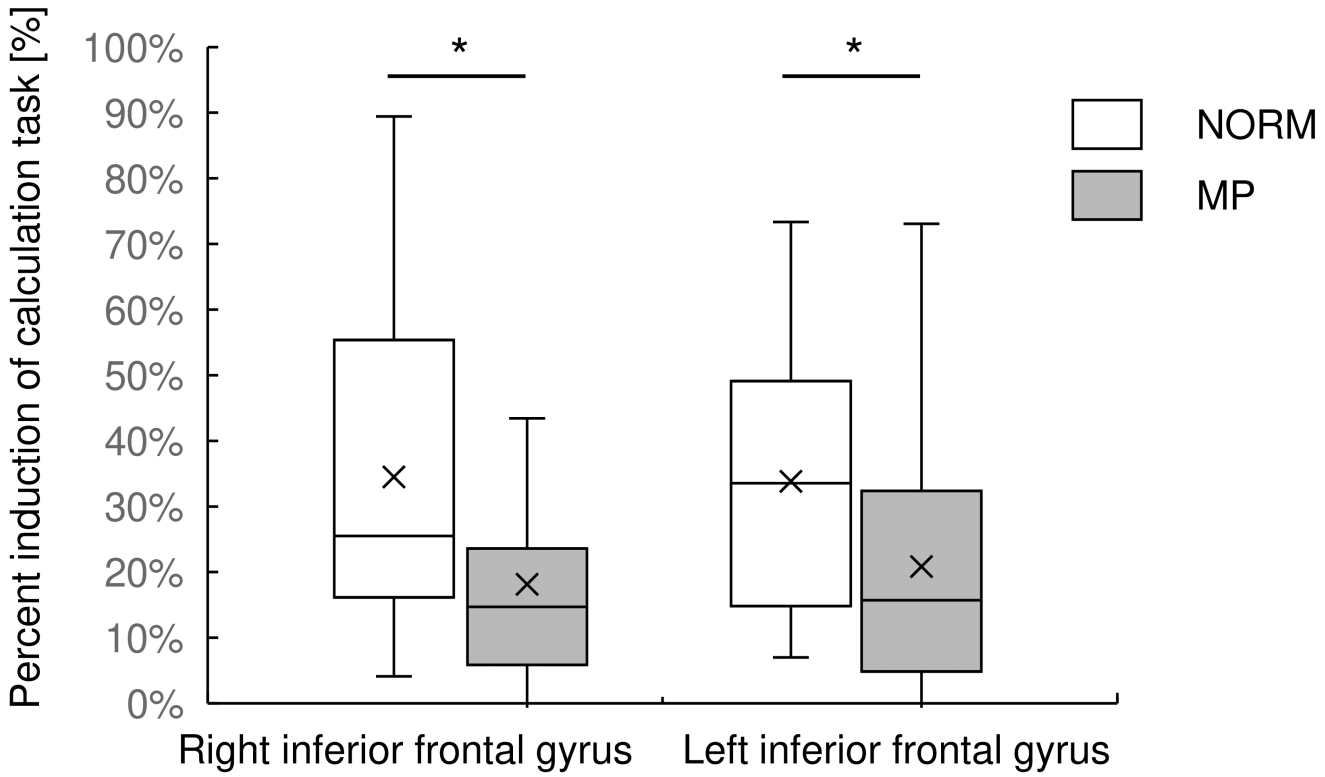

Figure 1

Chewing-induced BBF in NORM and MP.

Comparison of oxidized hemoglobin (Oxy-Hb) values in the bilateral inferior frontal gyrus during chewing task in participants with normal occlusion (NORM) and patient with mandibular prognathism (MP). The figure shows percent induction of BBF in chewing task normalized by calculation task (%). Box plots represent median (horizontal line), mean (´), 25th and 75th percentiles (boxes), and minimum and maximum values (whiskers). Right and left inferior frontal gyrus in each group are presented. NORM values (n=17) are shown as open boxes, and MP values (n=44) are shown as gray boxes. * $p < 0.05$ between groups (Mann–Whitney U test).

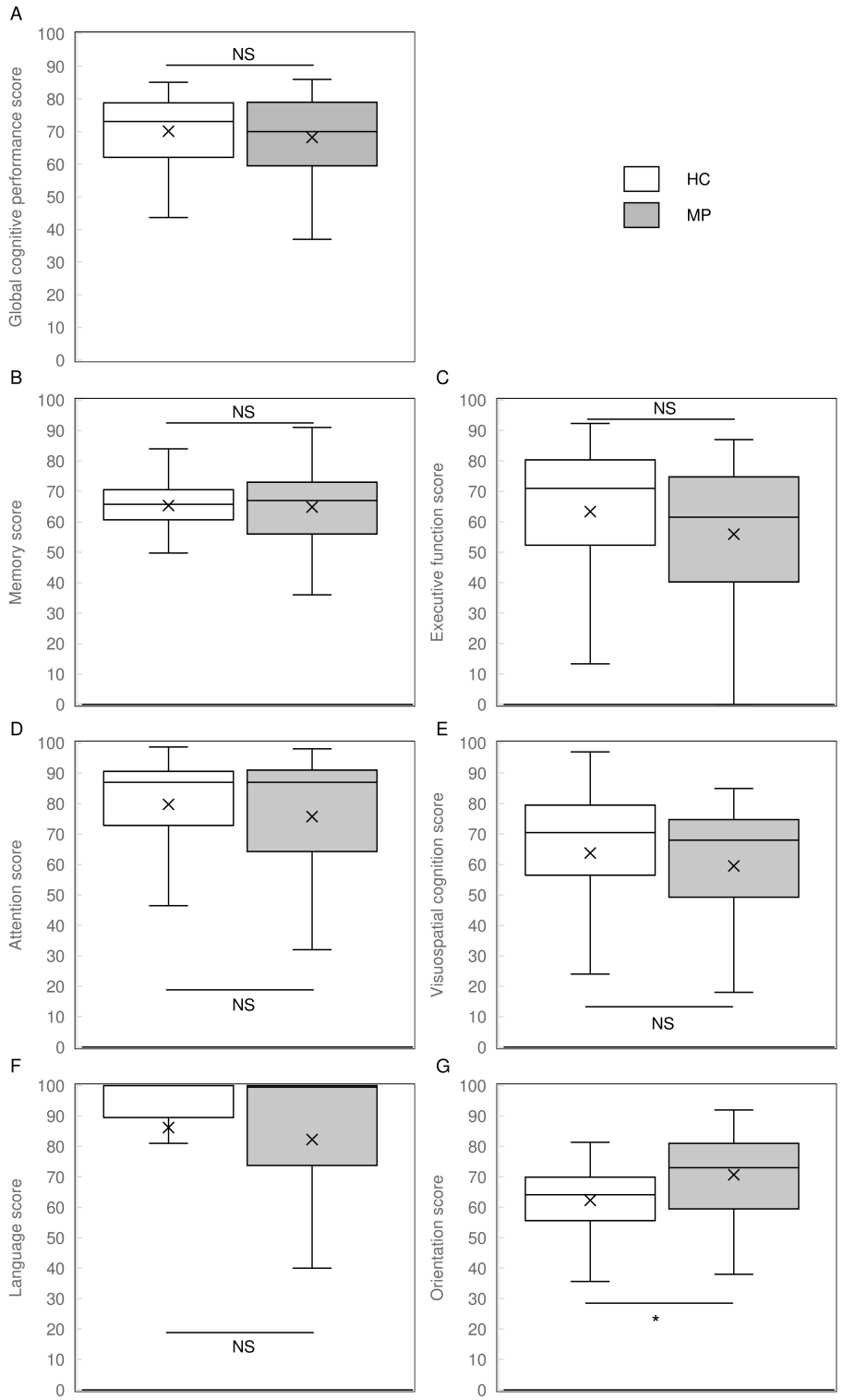

Figure 2

Comparison of cognitive function scores between HC and MP.

Cognitive function scores measured by Mirudake are shown. Box plots represent median (horizontal line), mean (´), 25th and 75th percentiles (boxes), and minimum and maximum values (whiskers). Global cognitive performance score (A), memory (B), executive function (C), attention (D), visuospatial cognition (E), language (F), and orientation (G) are shown, respectively. HC values (n=59) are shown as open boxes, and MP values

(n=44) are shown as gray boxes. NS: no statistically significant difference. * p < 0.05 between groups (Mann–Whitney U test).

Spearman	Global cognitive performance score	Memory	Executive function	Attention	Visuospatial cognition	Language	Orientation
Global cognitive performance score	1.00	0.74	0.62	0.52	0.40	0.49	0.49
Memory		1.00	0.28	NS	0.30	0.55	NS
Executive function			1.00	NS	NS	NS	0.39
Attention				1.00	NS	NS	NS
Visuospatial cognition					1.00	NS	NS
Language						1.00	NS
Orientation							1.00

Pearson / Bootstrap	Global cognitive performance score	Memory	Executive function	Attention	Visuospatial cognition	Language	Orientation
Global cognitive performance score	1.00	0.75	0.66	0.47	0.61	0.58	0.36
Memory		1.00	0.36	NS	0.53	0.46	NS
Executive function			1.00	NS	0.32	NS	0.36
Attention				1.00	NS	NS	NS
Visuospatial cognition					1.00	NS	NS
Language						1.00	NS
Orientation							1.00

0.0 – 0.2
0.2 – 0.4
0.4 – 0.7
0.7 – 1.0

Figure 3

Heatmap visualization of the correlation among cognitive function scores.

The degree of correlation between the scores in Spearman (upper panel) and Pearson with bootstrap resampling (1000 iterations, lower panel) are shown. Heatmap color coding indicates correlation strength: 0.0-

0.2 (weak correlation, white), 0.2-0.4 (weak to moderate correlation, blue), 0.4-0.7 (moderate to strong correlation, green), and 0.7-1.0 (strong correlation, red). NS: no statistically significant correlation.

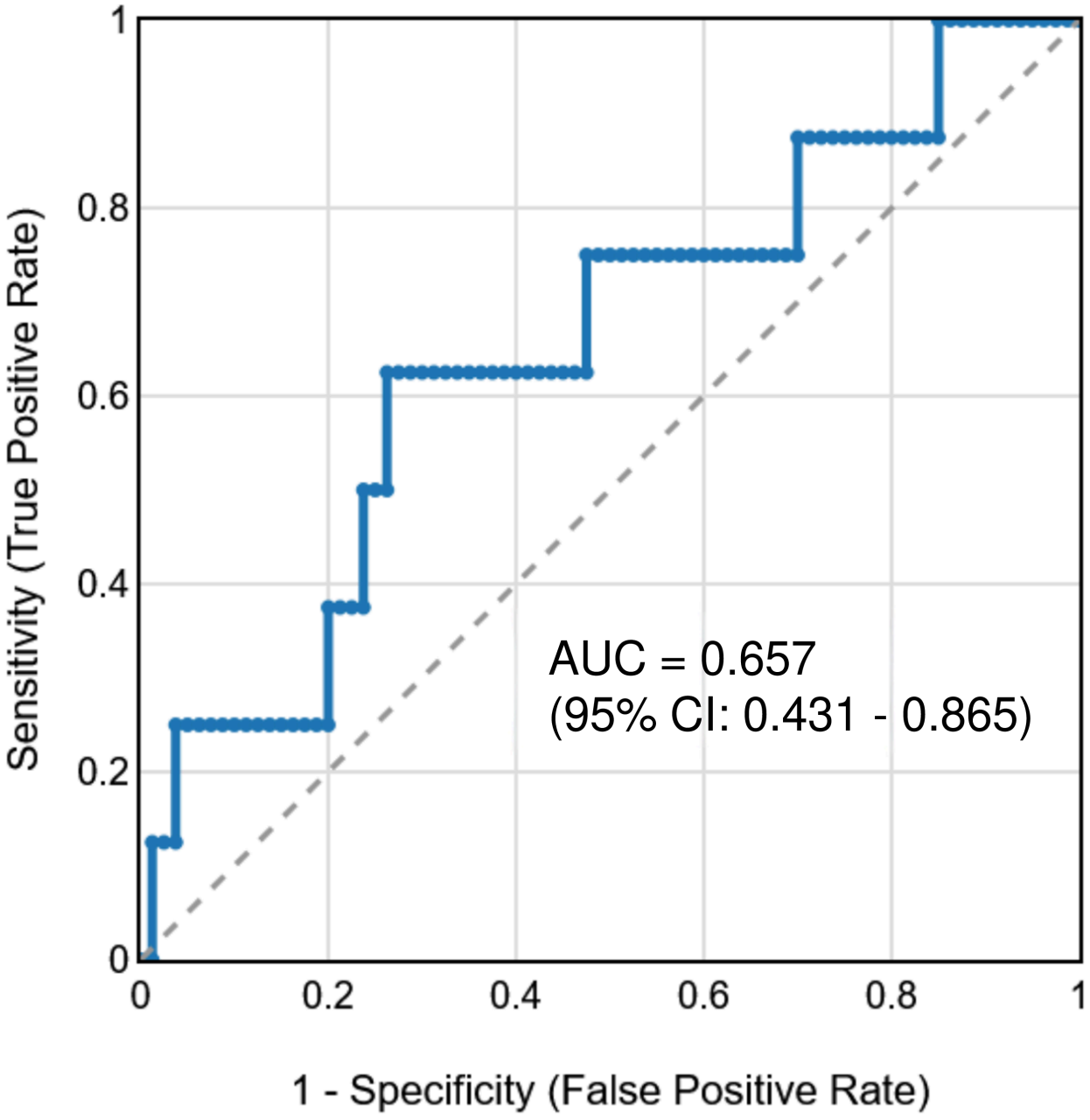

Figure 4

ROC curve analysis to evaluate the discriminative ability of BBF measurements for cognitive status classification.

BBF data from both left and right sides were pooled, resulting in a total sample size of 88 measurements (n = 88). The threshold was set as the Mirudake global cognitive performance score 50, which corresponding to the lower limit of the normal range in MMSE (i.e., MMSE ≥ 26), representing the boundary between normal cognitive

function and cognitive impairment. This classification gave 8 values below threshold and 80 values above threshold. ROC curve showing diagnostic performance of BBF biomarker. AUC = 0.657 (95% CI: 0.431-0.865, calculated with bootstrap resampling of 1000 iterations). Optimal threshold determined by Youden's index.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- Table1Equivalencetestresultssummary.docx
- Table2BilateralBBFcognitivefunctioncorrelation.docx
- Table3.PerformanceatSelectedThresholds.docx
- Table4AUC.docx